

AV NETWORK DESIGN REPORT

V18 M4500 spine and leaf, 10G fiber

October 1, 2026

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NOT VALIDATED

Estimate for planning only. Send this design to the Pro AV Design team for validation before ordering.

240

DEVICES

8

SWITCHES

0 W

POE LOAD

3.2 kW

EST. POWER

8.0 U

RACK SPACE

Networking engineered for AV over IP

NETGEAR AV switches, routers and WiFi access points are built specifically for networked media and AV, with out-of-the-box performance and easy setup for even the most complex installation. NETGEAR works with AV manufacturers and integration partners around the world, and supports the partners and associations that make the Pro AV and broadcast industries work.

● Optimized for AV over IP

Low-latency switching for audio, video and control, with PoE+ (30 W) and Ultra PoE++ (90 W) per port options.

● Certified AV profiles

NETGEAR AV OS with configuration profiles certified by leading AV manufacturers (Audinate, Crestron, Q-SYS, SMPTE ST 2110, NDI).

● Multicast that just works

NETGEAR IGMP Plus stops multicast flooding automatically, and Auto-LAG and Auto-Trunk connect switches without manual setup.

● Free central management

The NETGEAR Engage Controller manages switches, routers and WiFi 7 access points from one screen, and can set up the network before the hardware arrives.

● Scales with the project

From a single rack to redundant, active-active cores with 10G, 25G and 100G uplinks.

● Lower total cost

Energy-efficient hardware, a lifetime warranty on managed switches and less need for specialist IT resources.

IN THIS DESIGN

This design uses M4500 switches, all managed for free from one screen in the NETGEAR Engage Controller with NETGEAR AV OS.

BUILT FOR EVERY SPACE

Broadcast and live events · Corporate and commercial AV · Higher education · Medical · Unified communications · High-end residential

HELP FROM DESIGN TO DEPLOYMENT

NETGEAR has a dedicated Pro AV and broadcast network design team, pre-sales support and technical assistance. Free on-demand training and certification at academy.netgear.com covers networking basics through in-depth troubleshooting, and the top courses are AVIXA-accredited for CTS renewal units.

Support and services

Every NETGEAR managed switch includes a lifetime hardware warranty and complimentary support to get you started. Premium Support tiers add guaranteed response times, longer support hours and faster hardware replacement, so you can match support to how critical the system is.

	Warranty (included)	Sprint	Overdrive	Fastlane
Complimentary setup support	✓	✓	✓	✓
AI-driven chat support	✓	✓	✓	✓
Direct access to support engineers		✓	✓	✓
Support hours		24/5	24/7	24/7
Guaranteed response time (P1)		8 hours	2 hours	Custom
Priority case handling			✓	✓
Escalation to senior engineers			Level 2	Level 3
Hardware replacement	Fast RMA	Advance replacement	Next business day	Custom
Scheduled support calls			✓	✓
Operational reports			Quarterly	Custom
Designated or dedicated engineer				✓
Custom workflows and SLAs				✓

Managed and enterprise products include 90 days of complimentary phone, chat and online support. Fully managed switches include 3 years of Sprint Support. Sprint and Overdrive response times are engagement commitments for Priority 1 cases.

PROFESSIONAL SERVICES

Deployment services	Design validation, hands-on configuration and readiness testing, so the network is production-ready from day one. Remote, onsite or hybrid.
Health checks and optimization	Proactive network audits that find issues before they become outages, with a clear remediation plan.
Custom training and enablement	Training based on your deployed system: architecture walkthroughs, feature deep dives, best practices and troubleshooting.
Interoperability, staging and migration	Multi-vendor compatibility checks, staged configurations and low-risk migrations to NETGEAR.
Event services	Pre-event validation and real-time onsite or remote engineering support for live broadcasts, sporting events and other high-stakes shows.
Fastlane professional engagement	A high-touch engagement with senior engineering access for organizations that need deeper collaboration.
Part number	Service
PSP1104-10000S	On-site engineering assistance, full day
PRC0001-10000S	New remote configuration, full day

Pricing depends on project scope and the level of support, and includes travel and expenses. Ask your NETGEAR contact or distributor for a quote.

Design summary

Estimate for planning only. This bill of materials is generated automatically as a budgetary estimate. It must be validated by the ProAV Design team before ordering. Validation: proavdesign@netgear.com

ROOMS AND DEVICES

Device	Category	Qty	Link	PoE each
Main equipment room • MDF • 0 devices				
Leaf 1 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–
Leaf 2 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–
Leaf 3 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–
Leaf 4 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–
Leaf 5 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–
Leaf 6 • IDF, 100 m multimode fiber • 40 devices				
10G video endpoint (SFP+ fiber)	Video	40	10G Fiber	–

Network diagram



Green lines are uplinks, labelled with link count, speed and the optic or cable used. With a redundant core, every switch has its own full-capacity link group to each core, so either core can carry all traffic alone.

Bill of materials

Part number	Description	Qty	Where
SWITCHES			
CSM4532-100NAS	M4500-32C core switch	2	Main equipment room (core)
XSM4556-100NAS	M4500-48XF8C switch	6	Leaf 1, Leaf 2, Leaf 3, Leaf 4, Leaf 5, Leaf 6
OPTICS			
ACM761-10000S	100GBASE-SR4 QSFP28, multimode (MPO)	96	Leaf 1, Main equipment room (core), Leaf 2, Leaf 3, Leaf 4, Leaf 5, Leaf 6
AXM761-10000S	10GBASE-SR SFP+, multimode	240	Leaf 1, Leaf 2, Leaf 3, Leaf 4, Leaf 5, Leaf 6
CABLING			
ACC761-10000S	100G QSFP28 DAC 1m, passive copper	4	Main equipment room (core)

Part numbers are for Americas. Pricing and availability are provided by NETGEAR or your distributor after validation.

Power and rack space

Room	Est. draw	Datasheet max	Heat load	Current	Circuits	UPS size	Rack
Leaf 1	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Leaf 2	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Leaf 3	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Leaf 4	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Leaf 5	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Leaf 6	394 W	394 W	1,344 BTU/hr	3.6 A at 110 V	1 × 20 A	438 VA+	1.0 U
Main equipment room (core)	852 W	852 W	2,907 BTU/hr	7.7 A at 110 V	1 × 20 A	947 VA+	2.0 U

POWER SUPPLIES PER SWITCH

Room	Device	Qty	PoE load each	Power supplies	PoE available	Headroom	Est. draw each
Leaf 1	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Leaf 2	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Leaf 3	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Leaf 4	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Leaf 5	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Leaf 6	M4500-48XF8C	1	–	2x APS750W (1+1, included)	–	–	394 W
Main equipment room (core)	M4500-32C	2	–	2x APS750W (1+1,	–	–	426 W

Room	Device	Qty	PoE load each	Power supplies	PoE available	Headroom	Est. draw each
included)							

Estimated draw = datasheet maximum without PoE + PoE load ÷ 0.9 (PSU efficiency). Circuits assume 80% continuous load.

Checks

NOTE	Inter-core link sized for largest single switch (one uplink group fails), and never below the busiest single switch: needs 400 Gbps, provided $4 \times 100G = 400$ Gbps.
NOTE	Switches interconnect with LAG uplinks and IGMP Plus (no stacking), which keeps AVB and PTP available.
NOTE	Uplink LAGs use 1, 2, 4 or 8 links to each core, and each core link group carries the switch's full load on its own so one core can fail (sizes from Tool_Settings: 1, 2, 4, 8), so LAG hashing spreads traffic evenly.

UPLINK PLAN

Room	Switch	Units	Uplinks each	Speed	Run	Optic / cable
Leaf 1	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S
Leaf 2	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S
Leaf 3	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S
Leaf 4	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S
Leaf 5	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S
Leaf 6	M4500-48XF8C	1	8	100G	MMF, 100 m	ACM761-10000S